Lab 05 Manual

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SRT411NAA: Digital Data Analytics

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Setting Up ELK Stack (Task 01)

Install Elasticsearch

Commands followed:

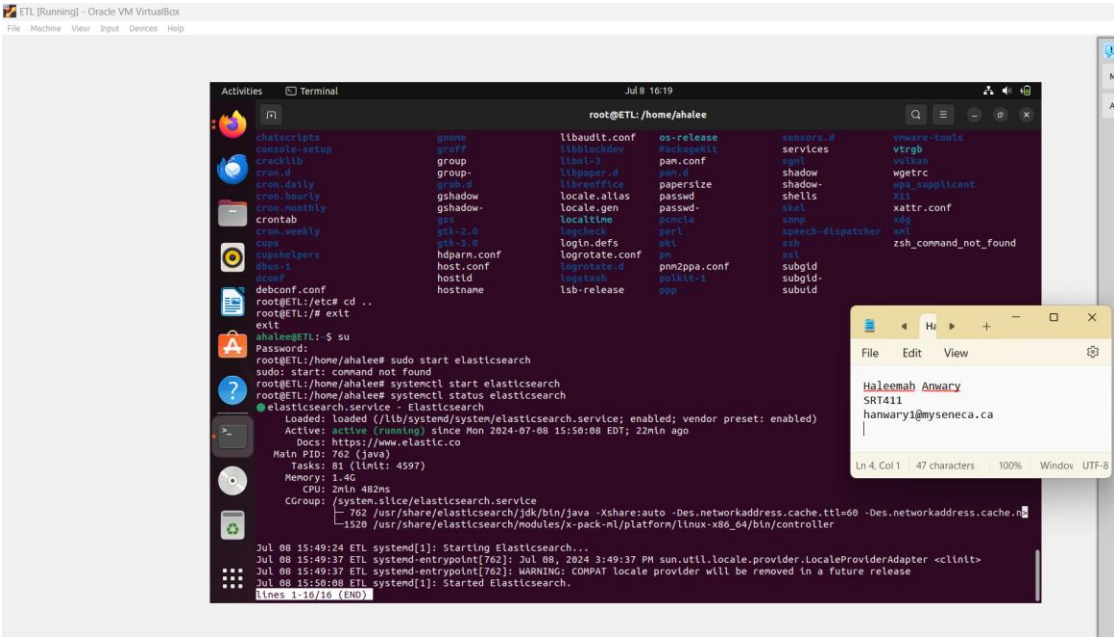
apt install default-jdk default-jre -y

curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | apt-key add -

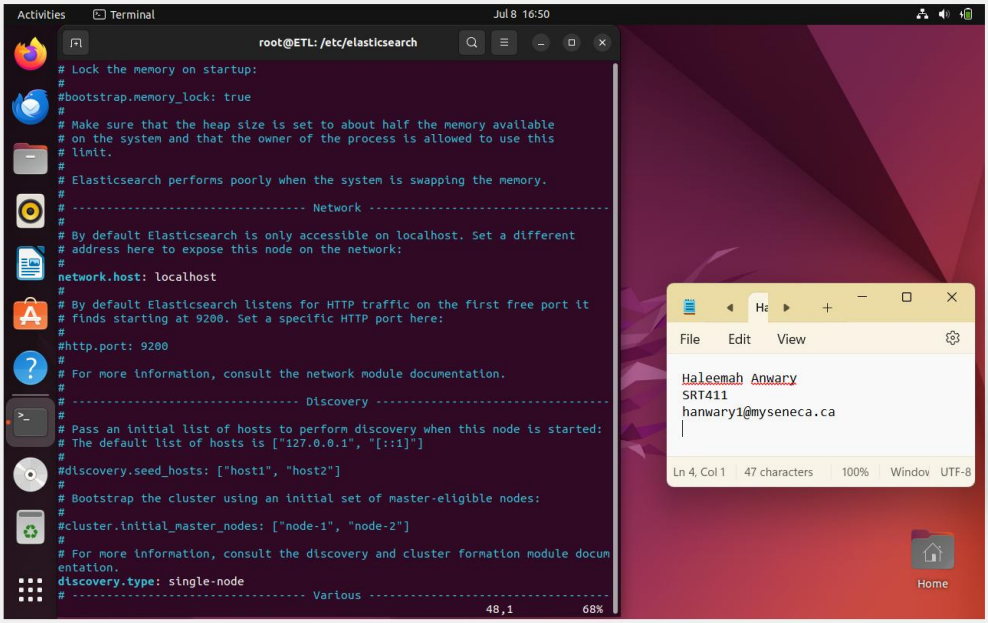
echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" > /etc/apt/sources.list.d/elastic-7.x.list

apt update

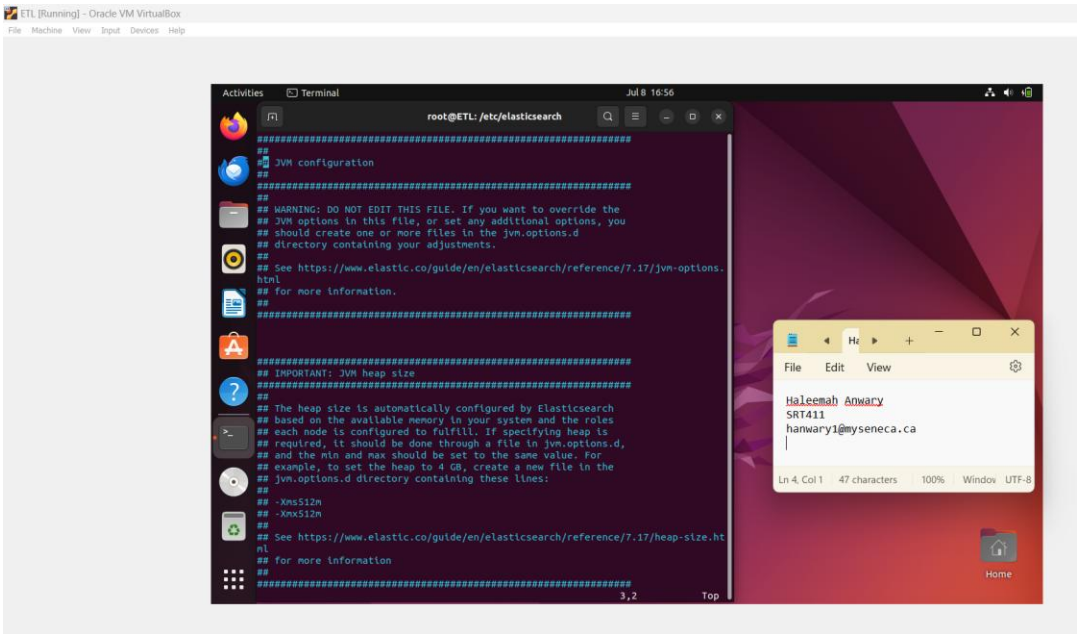
apt install elasticsearch -y



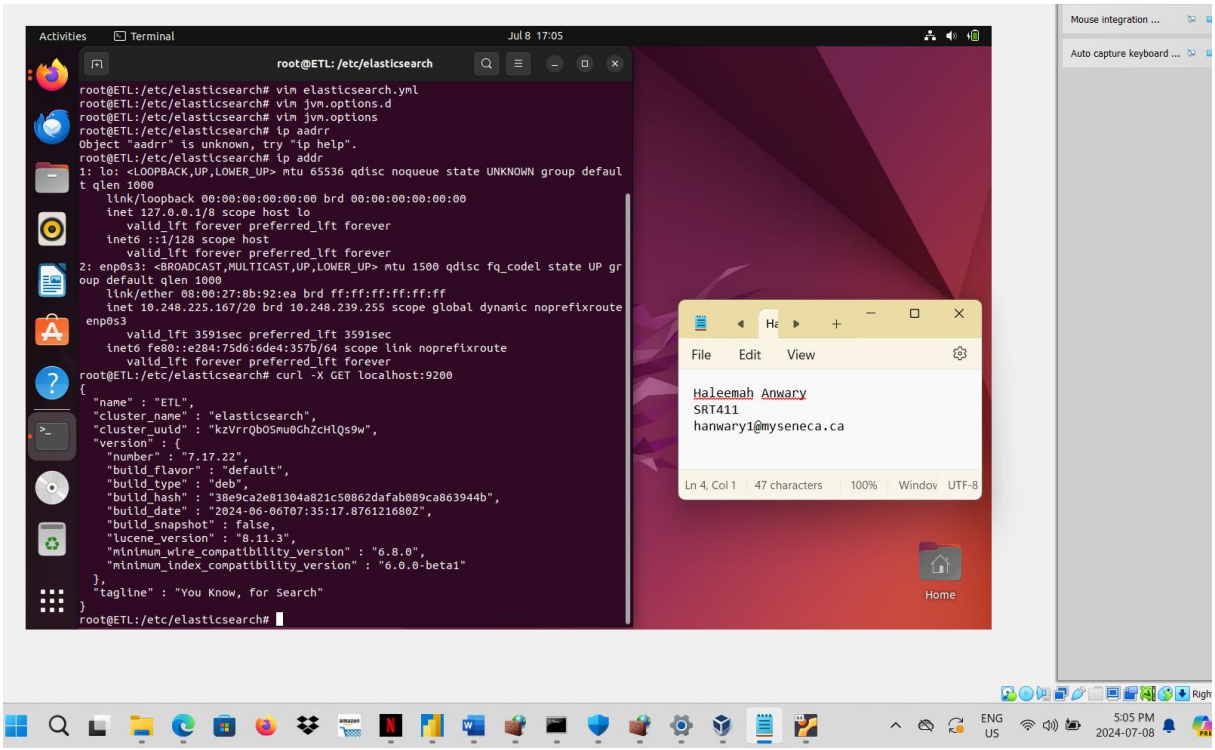
vim /etc/elasticsearch/elasticsearch.yml



Configure the JVM heap memory

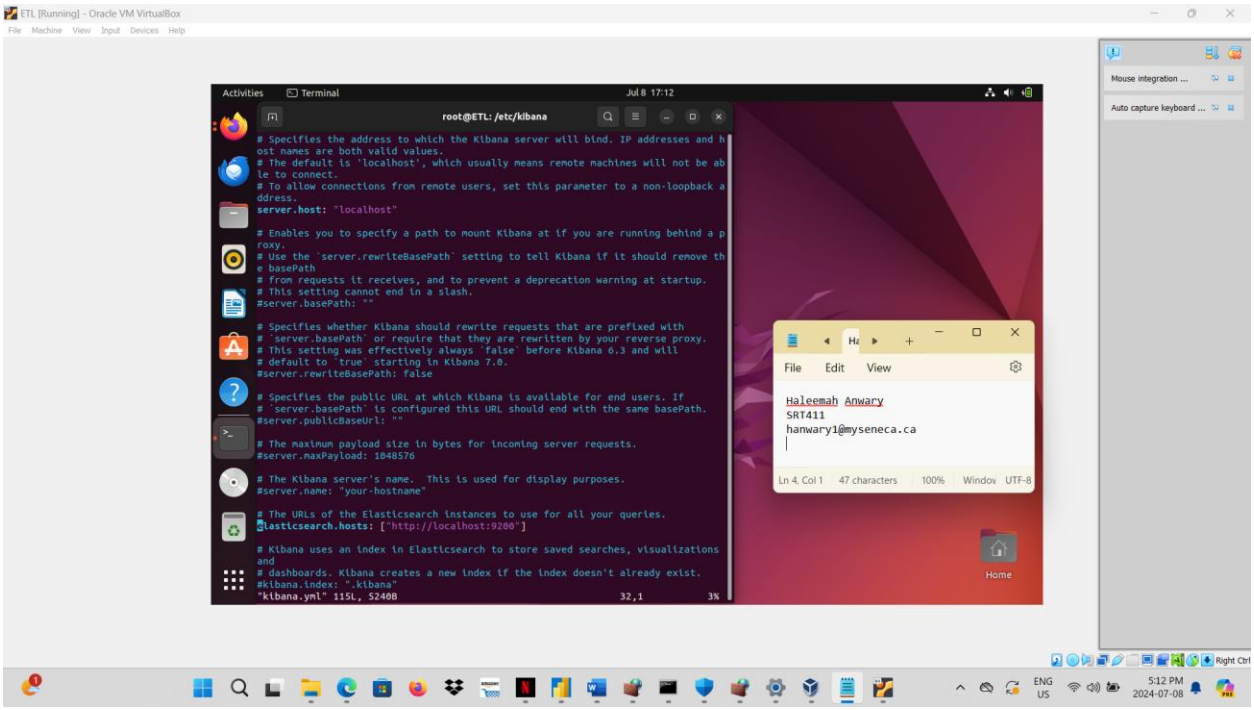


Verify the working of elasticsearch

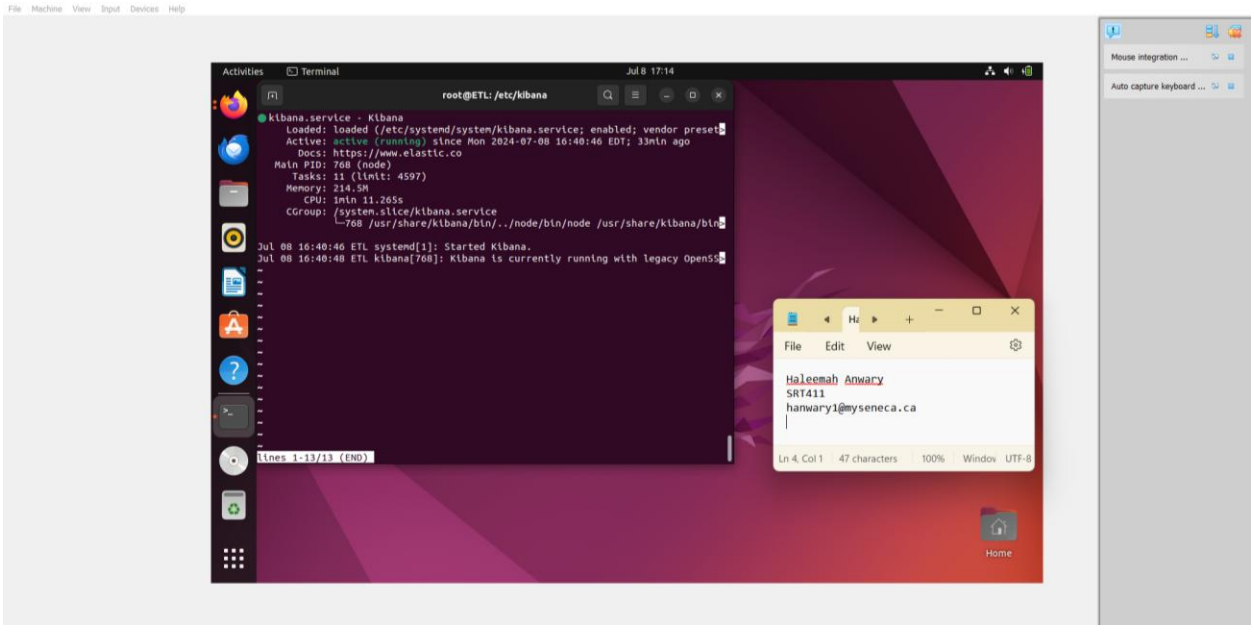


Install Kibana

vim /etc/kibana/kibana.yml



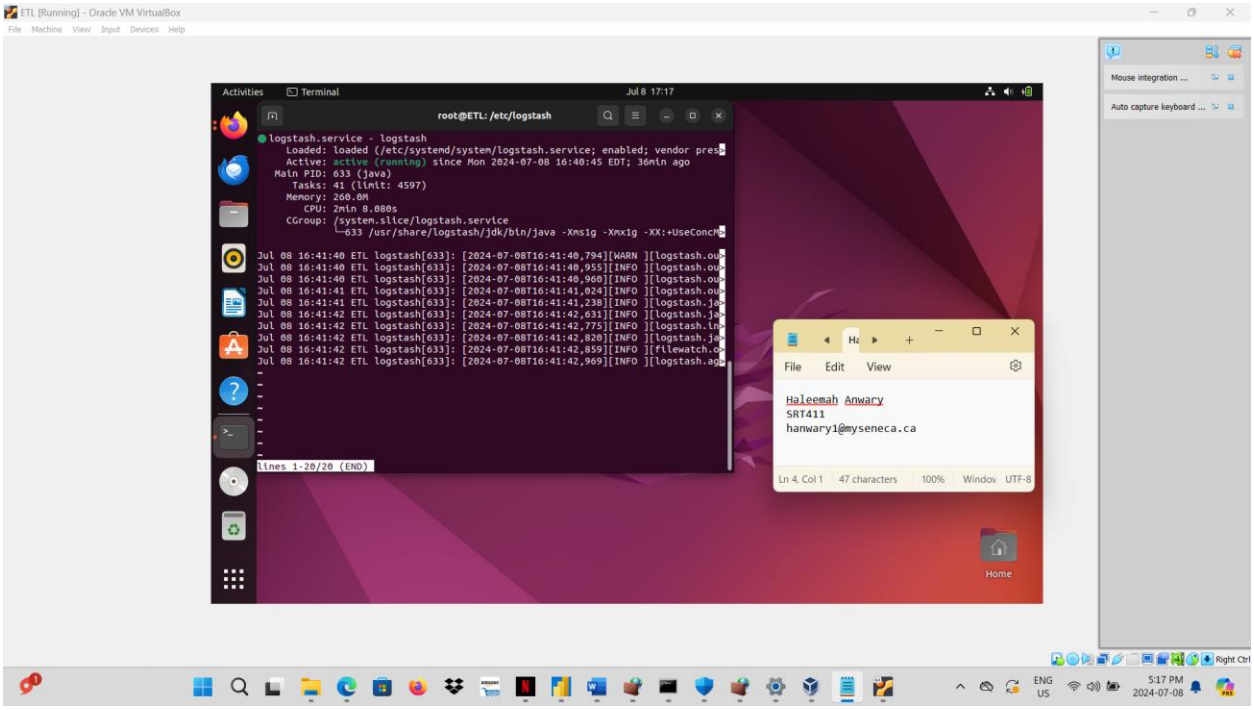
Verify Kibana Install



Install Logstash

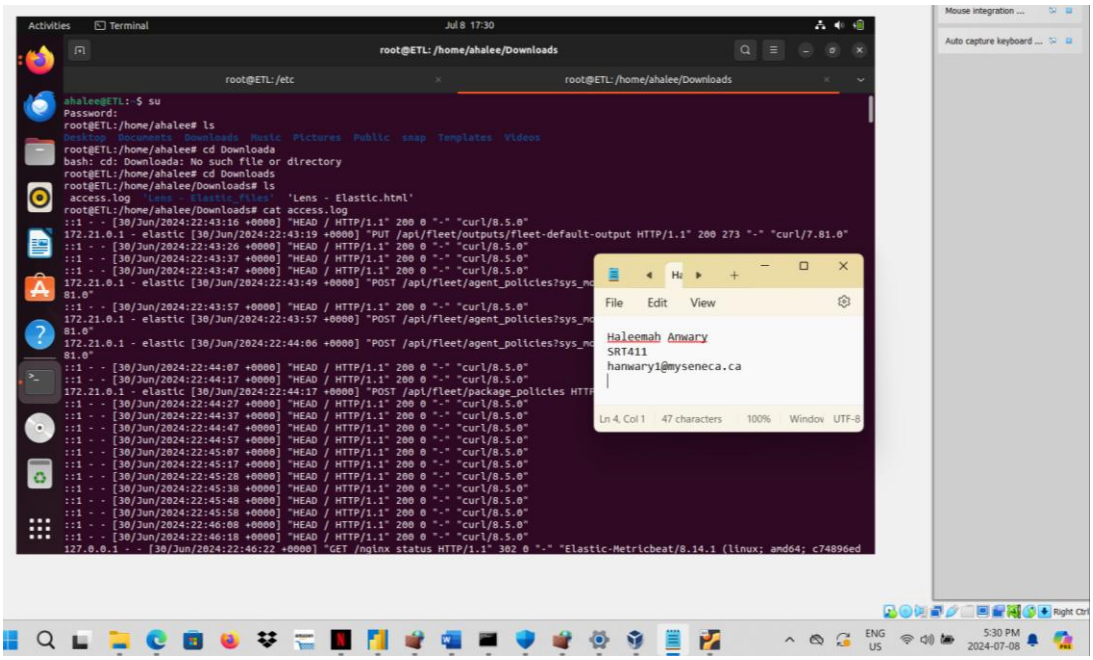
apt install logstas

systemctl start logstash
systemctl enable logstash

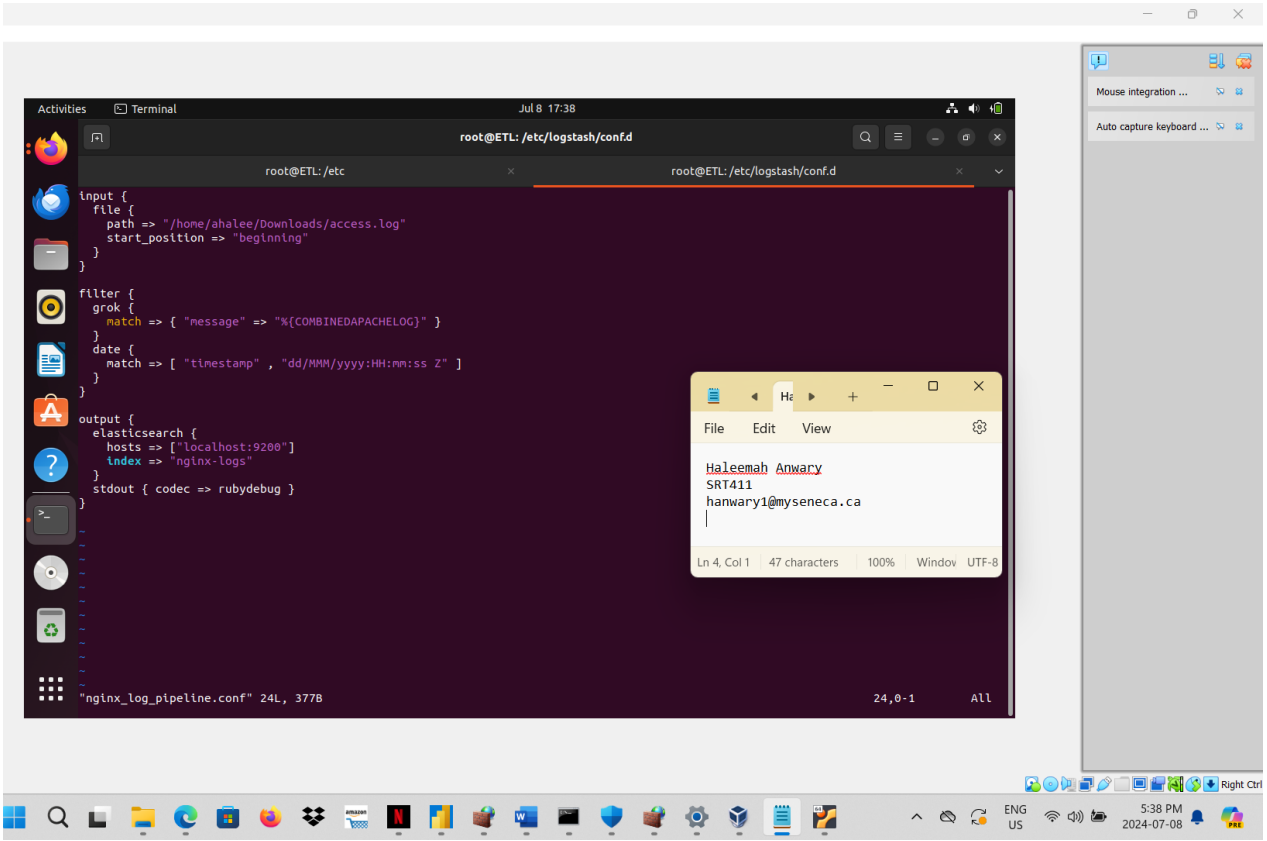


Ingesting Nginx Log (Task 02)

Access.log file placed in the Downloads directory

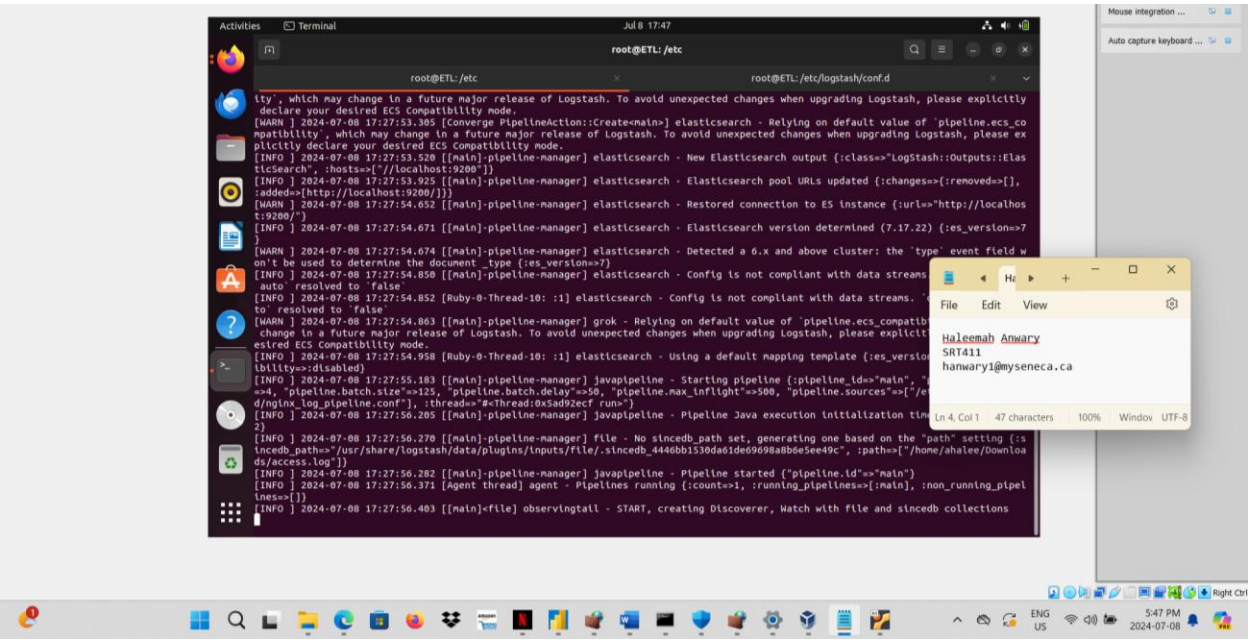


Write a logstash pipeline to ingest this nginx access.log

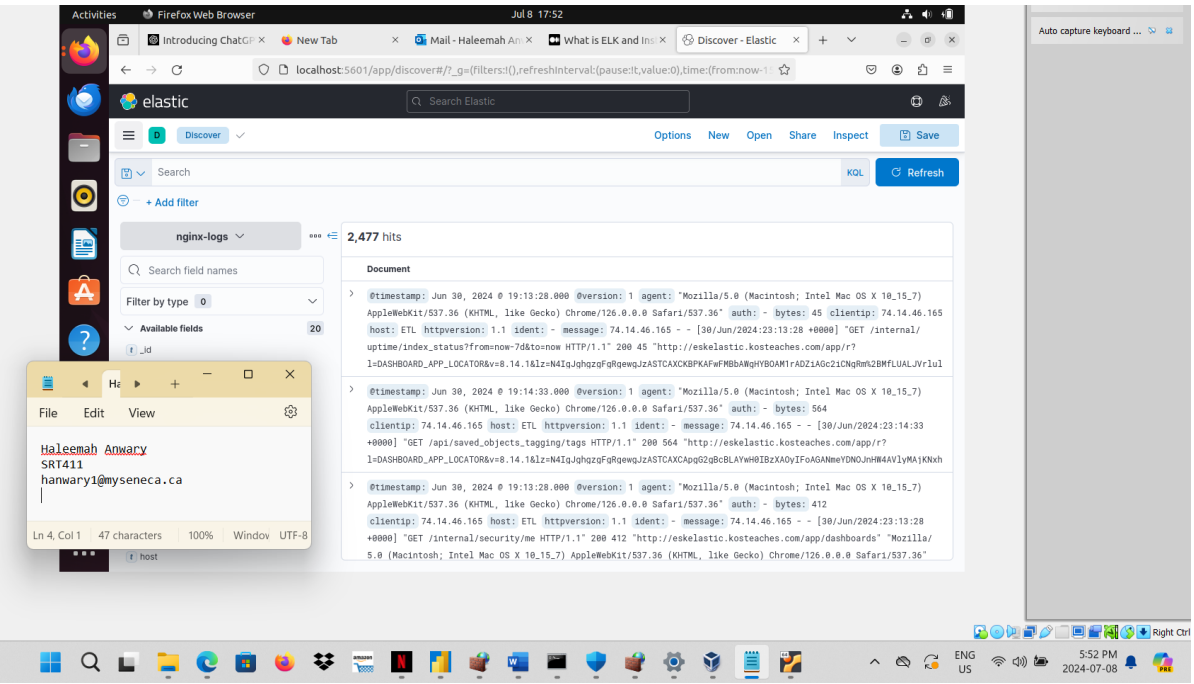


Ingest the log lines into ELK stack

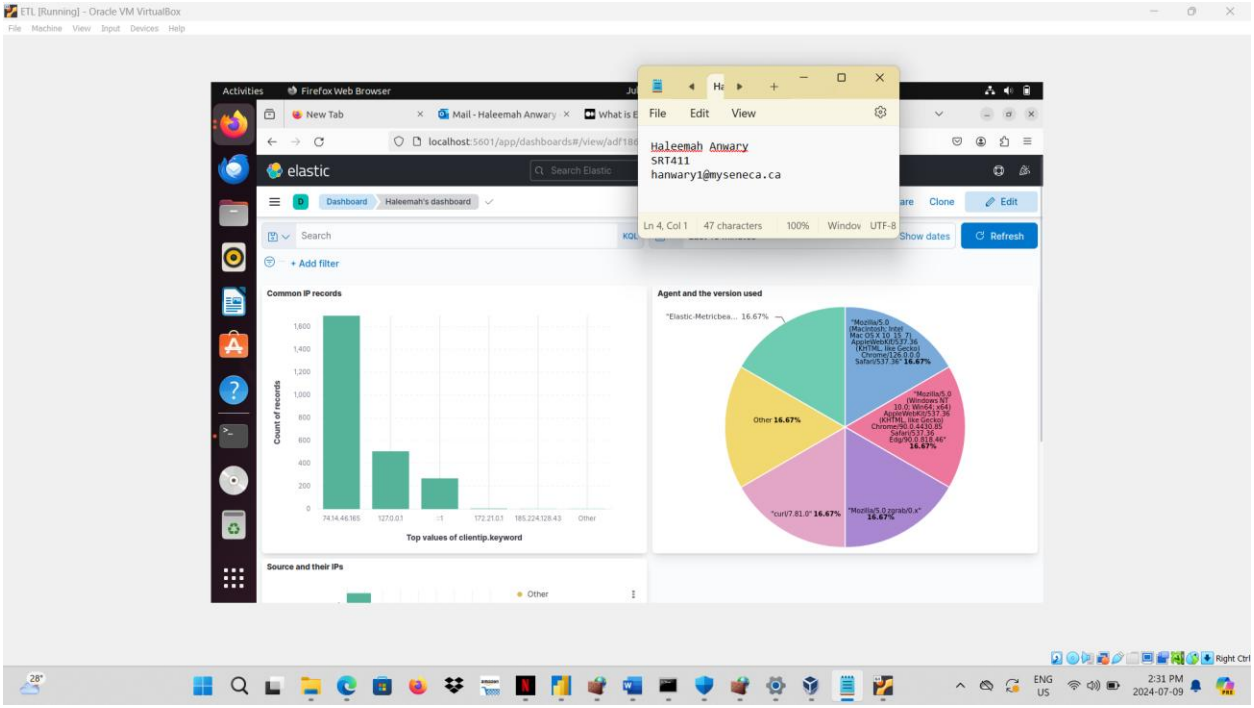
sudo /usr/share/logstash/bin/logstash -f /etc/logstash/conf.d/nginx_log_pipeline.conf

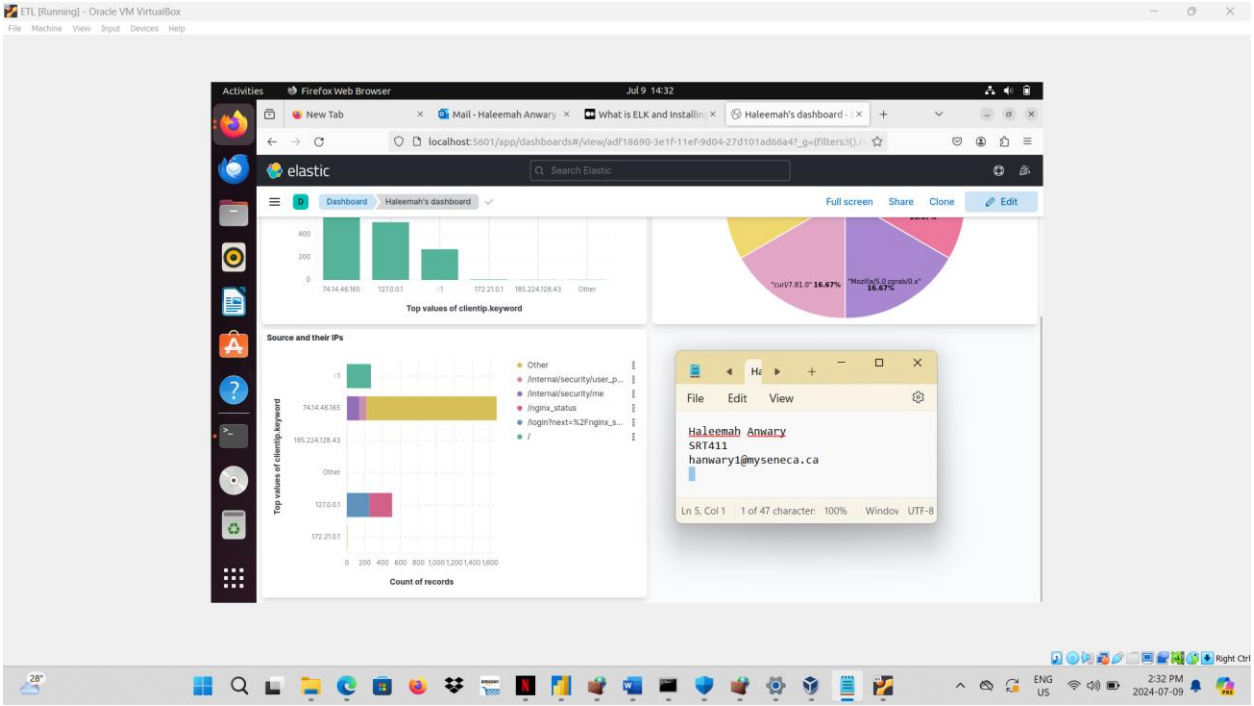


Show the documents in Discover



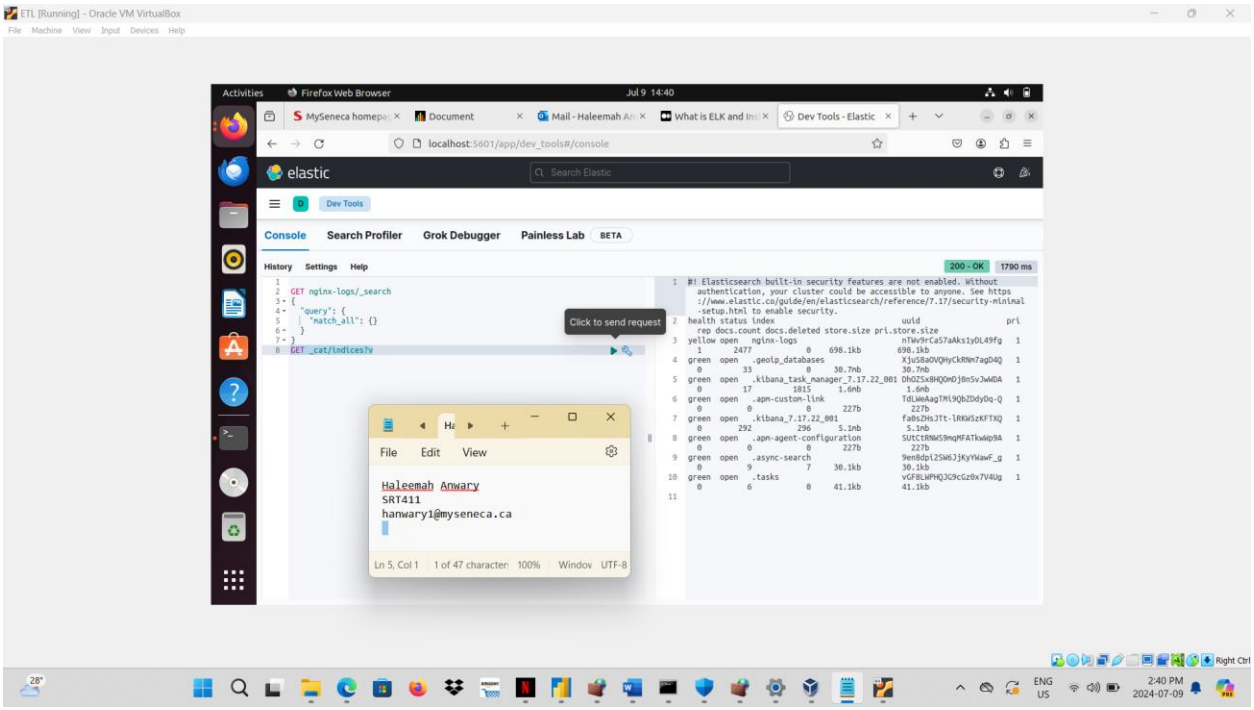
Build a dashboard like the Ingest Pipeline Dashboard shown in Nginx Dashboard Overview section above using the Logstash ingested documents.



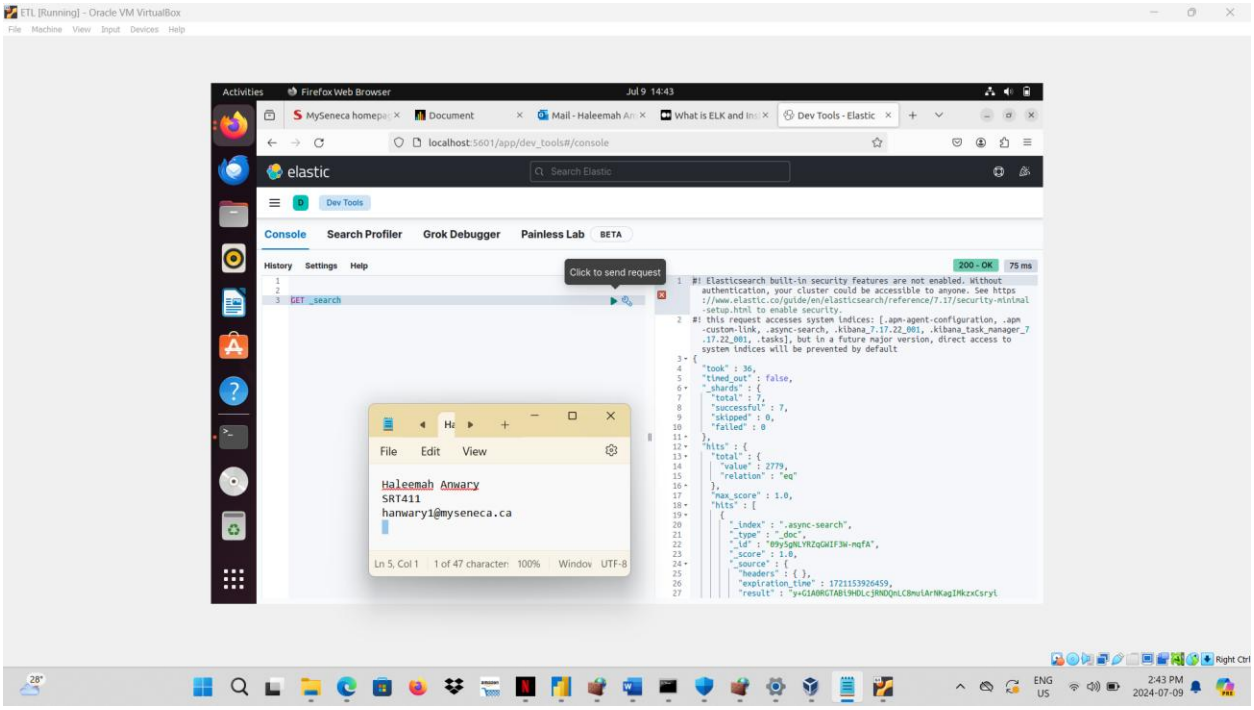


Running Queries in ELK Stack (Task 03)

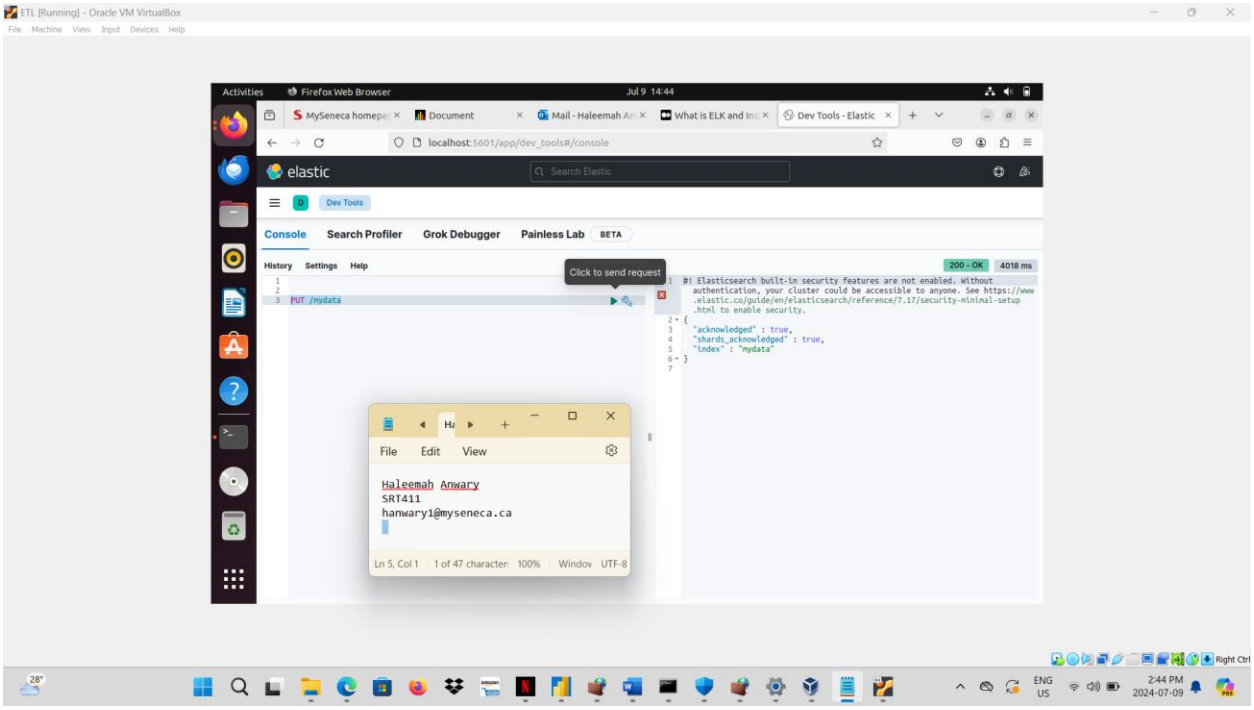
1) Query the number of indices in the cluster and their names



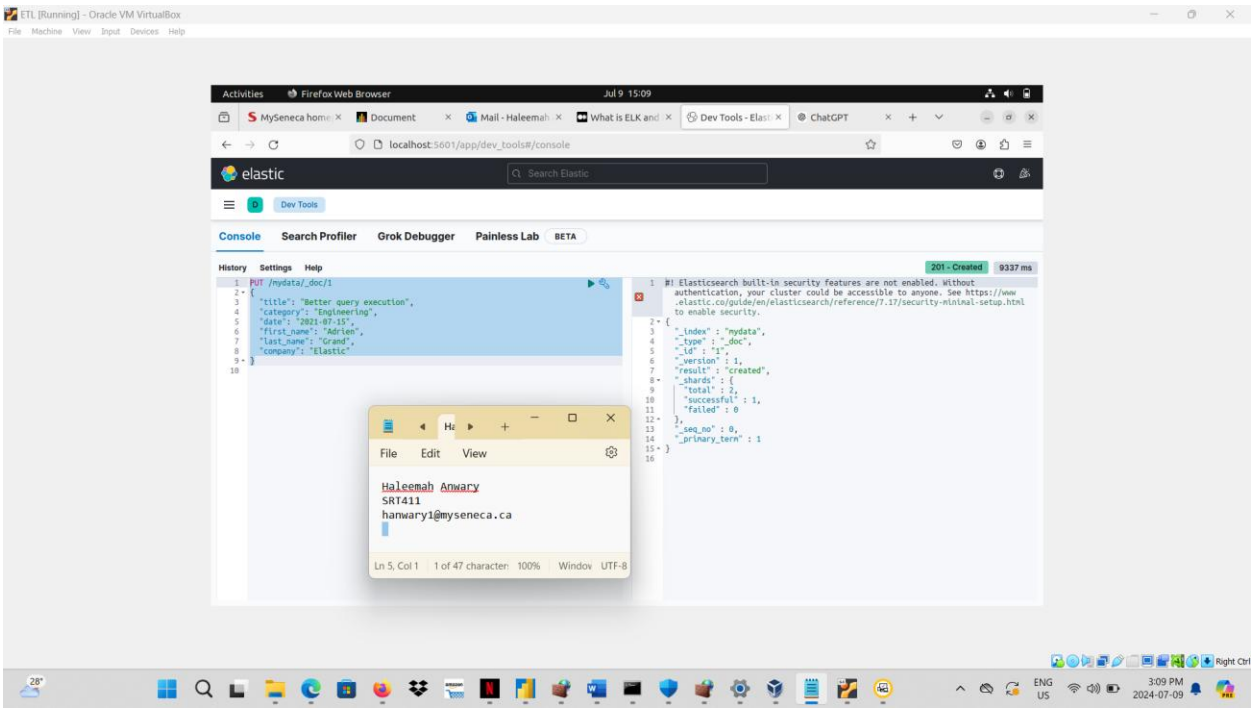
2) Query all documents in all indices of your cluster

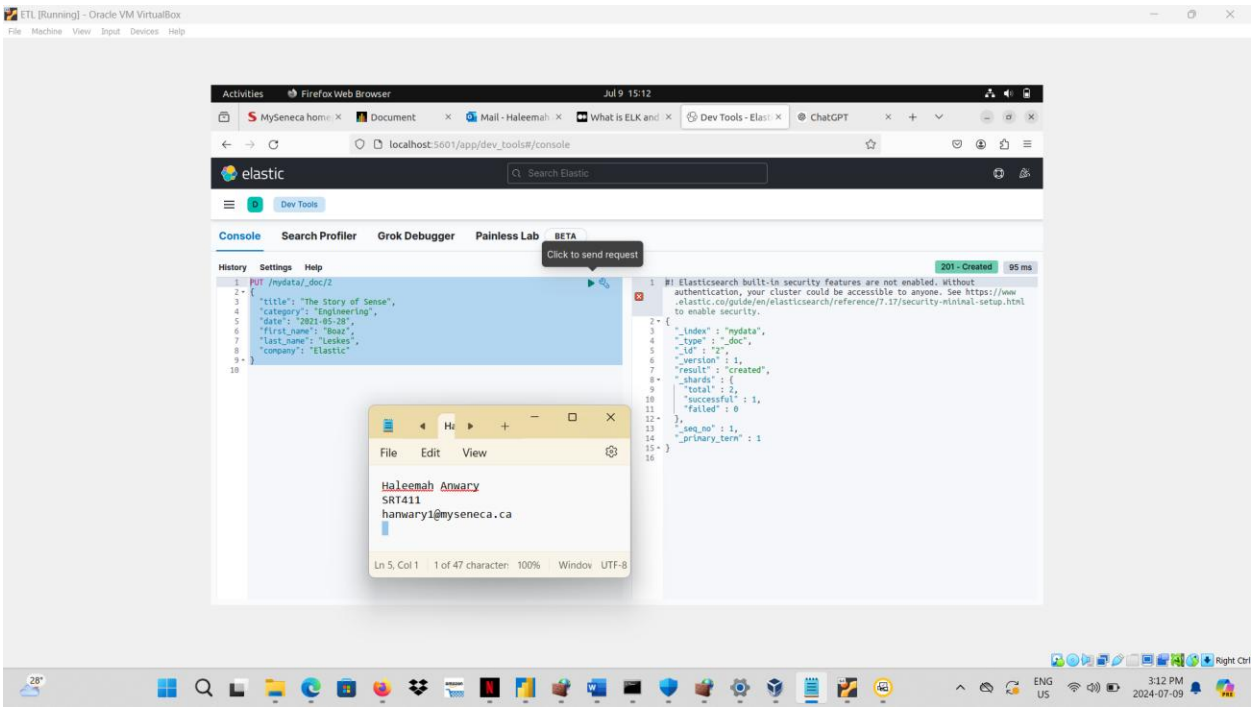


3) Create a new index named “mydata”

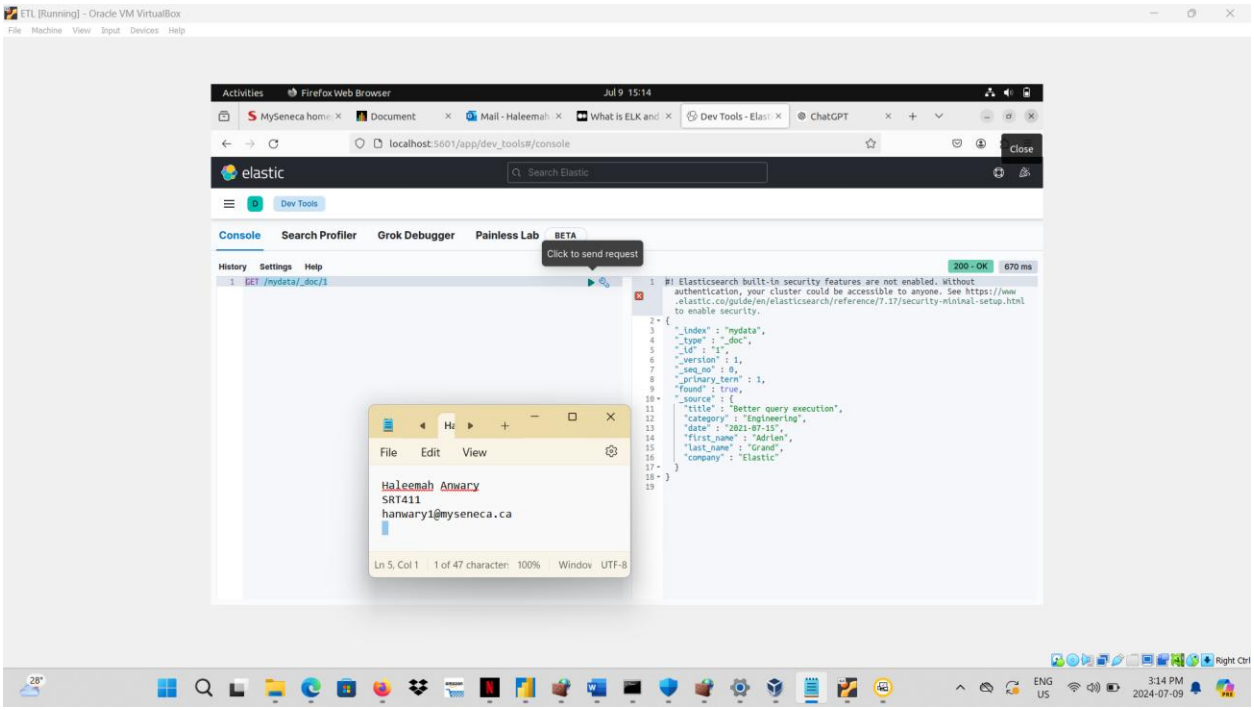


4) Add documents to the “mydata” index

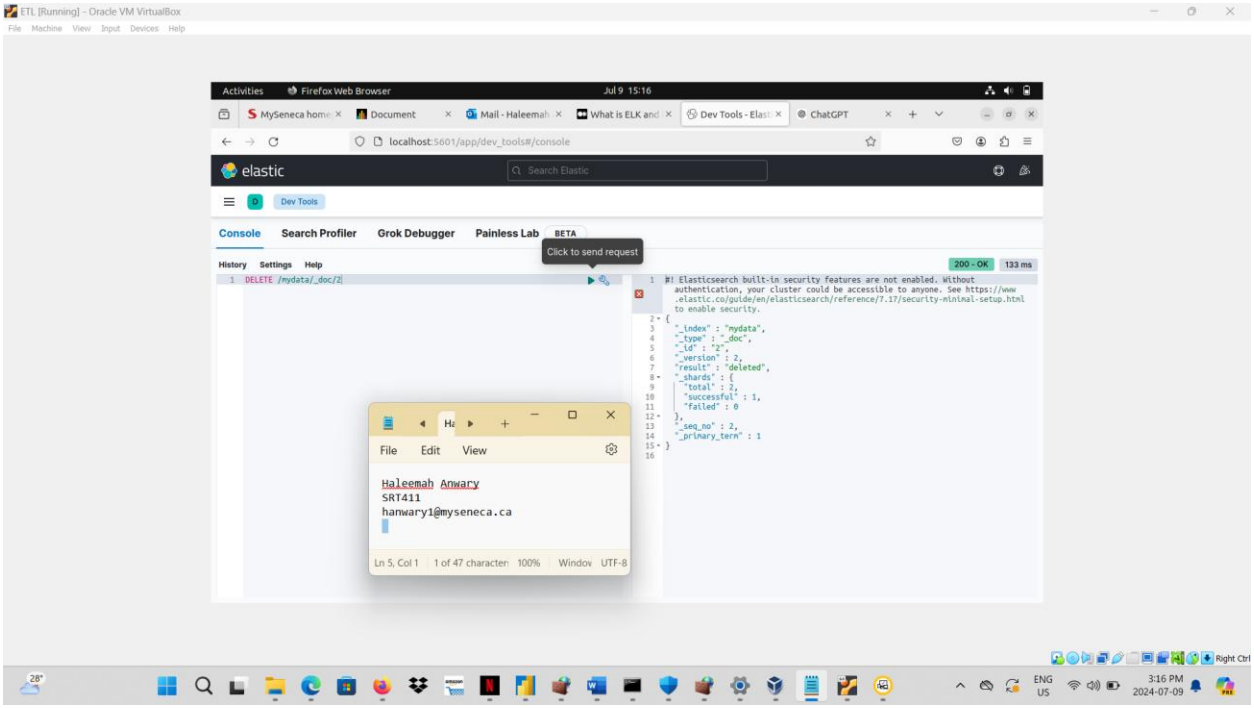




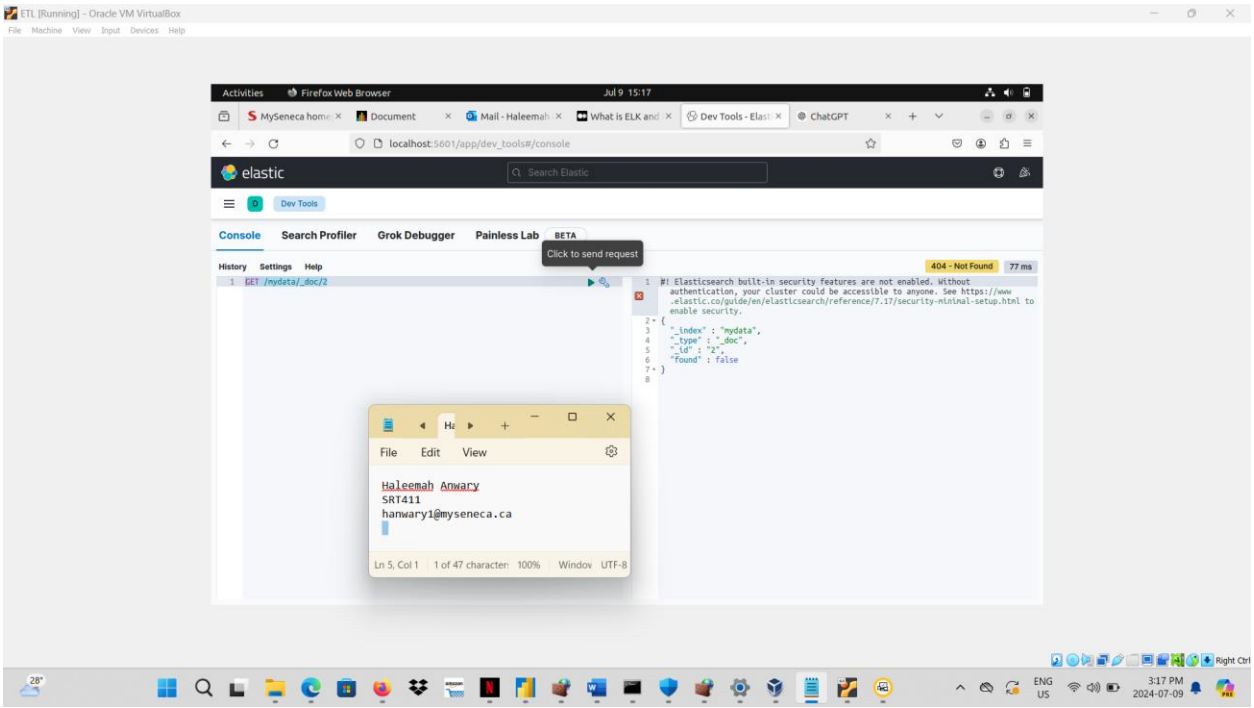
5) Retrieve the document with id 1 from the “mydata” index



6) **Delete the document with id 2 from the “mydata” index**



7) **Verify the deletion of the document with id 2**



Learning Reflection (Task 04)

1. What are clusters and nodes in Elasticsearch?

In Elasticsearch, a cluster is a collection of one or more nodes that together holds data and provides indexing and search capabilities. Nodes are individual instances of Elasticsearch that work together to form a cluster. Each node can be configured to perform different roles within the cluster.

2. What are the main differences between master nodes and data nodes in Elasticsearch?

Master nodes are responsible for cluster-wide actions such as creating or deleting indices, adding or removing nodes, and rebalancing shards. Data nodes, on the other hand, are responsible for storing and executing data-related operations such as CRUD operations, search, and aggregations.

3. Which types of nodes in Elasticsearch hold the documents?

Data nodes in Elasticsearch are the types of nodes that hold the actual documents and execute data-related operations. They store, index, and search data within the cluster.

4. How does Elasticsearch store data?

Elasticsearch stores data in a structure called an inverted index. This allows for fast full-text searches and efficient querying. Data is stored as JSON documents, and each document is stored within an index, which is further divided into shards for scalability and performance

5. What are the benefits of using JSON objects in Elasticsearch?

JSON objects are used in Elasticsearch to represent data because of their flexibility, simplicity, and compatibility with various programming languages. They allow for easy representation of complex hierarchical data structures and are natively supported by Elasticsearch's RESTful API.

6. How does data storage in Elasticsearch compare to traditional relational database management systems (RDBMS)?

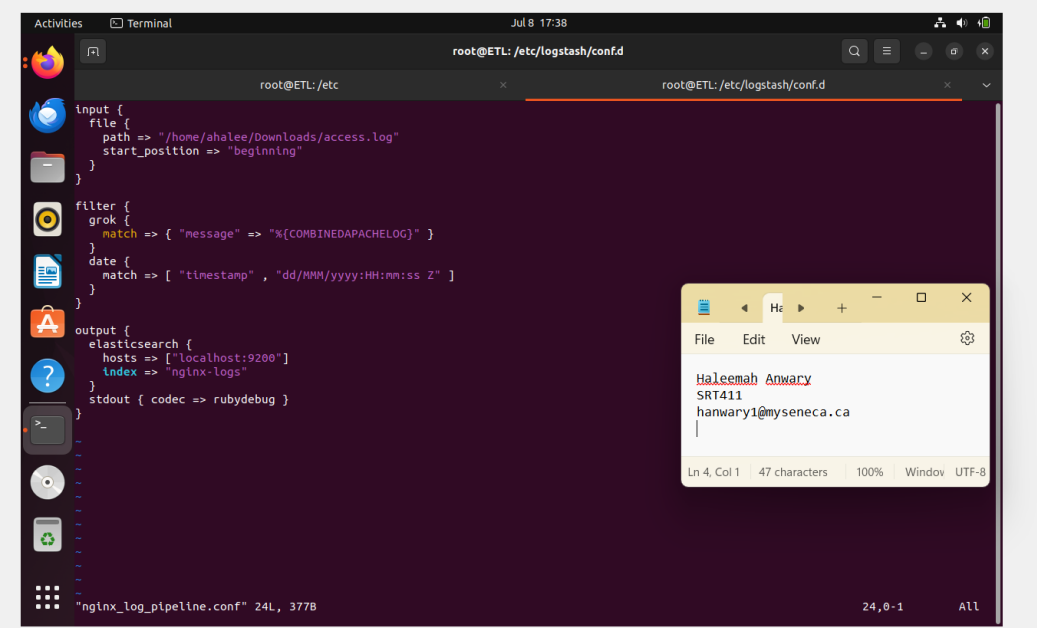
In Elasticsearch, data storage is schema-less, allowing for flexible and dynamic data modeling. This differs from traditional relational database management systems (RDBMS), which require a predefined schema and are less flexible when it comes to handling unstructured or semi-structured data. Additionally, Elasticsearch's distributed architecture and horizontal scalability provide better performance for search and analytics use cases compared to traditional RDBMS.

Set up Process:

Step 1) I followed the following website to download Elasticsearch, Logstash, and Kibana

What is ELK and Installing ELK stack (elasticsearch, logstash, kibana) in Ubuntu | by Cyber Tool Guardian | Medium

Step 2) I downloaded the access. Log file to my Downloads directory and wrote a logstash pipleline in the /etc/logstash/config.d path. I got this code from Chat



Step 3: I ran it with the command: sudo /usr/share/logstash/bin/logstash -f /etc/logstash/conf.d/nginx_log_pipeline.conf

Step 4) once I opened kibana on web, in the discovery section, this is what was loaded, I didn't have to do anything more to it.

